

# Supplementary Material for: When Accuracy Saturates: A Feature-Node Graph Representation for Deployable and Transportable Cardiovascular Risk Prediction

This file collects the full versions of four secondary robustness tables that are summarised in prose in the main text but omitted from the printed manuscript for length. No number here differs from what was used to support the corresponding claim in the main text; these tables provide the complete per-configuration detail underlying those summaries.

TABLE S1: CORRELATION-THRESHOLD SENSITIVITY

Referenced from the main text as “Threshold sensitivity” (Section 4.7 there). Full detail across  $\tau \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$ .

TABLE S2: CONVOLUTION-OPERATOR COMPARISON

Referenced from the main text as item (v) of the ablation study (“The choice of message-passing operator is not critical”). Full performance table and paired significance tests for GCN vs. GAT, GraphSAGE and GIN.

TABLE S3: IDENTITY-EMBEDDING DIMENSION SWEEP —  
PERFORMANCE

Referenced from the main text in “Does the reversal generalise beyond  $d = 8$ ?” Full performance and effective-rank table across  $d \in \{0, 2, 4, 8, 16, 32\}$  and three topologies.

TABLE S4: IDENTITY-EMBEDDING DIMENSION SWEEP — PAIRED  
GRAPH GAP

Companion to Table S3: the paired internal graph gap at each embedding dimension, Holm-corrected jointly across all twelve (dimension, contrast) hypotheses.

TABLE S1  
SENSITIVITY OF THE FULL MODEL TO THE PEARSON CORRELATION THRESHOLD  $\tau$ .

| tau  | Mean edges | Accuracy          | F1                | ROC-AUC           | MCC               |
|------|------------|-------------------|-------------------|-------------------|-------------------|
| 0.05 | 59.6       | 0.8087 +/- 0.0689 | 0.7938 +/- 0.0675 | 0.8997 +/- 0.0437 | 0.6215 +/- 0.1297 |
| 0.1  | 47.2       | 0.8087 +/- 0.0761 | 0.7991 +/- 0.0677 | 0.9032 +/- 0.0418 | 0.6255 +/- 0.1384 |
| 0.15 | 34.4       | 0.8109 +/- 0.0652 | 0.7971 +/- 0.0597 | 0.9051 +/- 0.0413 | 0.6280 +/- 0.1240 |
| 0.2  | 25.4       | 0.8076 +/- 0.0770 | 0.7979 +/- 0.0705 | 0.9035 +/- 0.0426 | 0.6241 +/- 0.1425 |
| 0.3  | 17.2       | 0.8064 +/- 0.0842 | 0.7894 +/- 0.0815 | 0.8973 +/- 0.0430 | 0.6201 +/- 0.1610 |

TABLE S2

CONVOLUTION-OPERATOR COMPARISON. PANEL (A) REPORTS PERFORMANCE FOR EACH OPERATOR; PANEL (B) REPORTS THE PAIRED SIGNIFICANCE TESTS AGAINST GCN. ALL  $p$ -VALUES IN PANEL (B) ARE MEDIANS OVER THE MATCHED RUNS, WITH HOLM CORRECTION APPLIED WITHIN THE FAMILY.

| (a) Performance by operator  |                   |                     |                     |                     |                     |                     |                     |                     |  |
|------------------------------|-------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|--|
| Operator                     | Params            | Accuracy            | Precision           | Recall              | F1                  | ROC-AUC             | MCC                 | Specificity         |  |
| GCN (Ours)                   | 1281              | 0.8109 $\pm$ 0.0652 | 0.8044 $\pm$ 0.0999 | 0.8011 $\pm$ 0.0748 | 0.7971 $\pm$ 0.0597 | 0.9051 $\pm$ 0.0413 | 0.6280 $\pm$ 0.1240 | 0.8196 $\pm$ 0.1208 |  |
| GAT                          | 1409              | 0.8109 $\pm$ 0.0718 | 0.8031 $\pm$ 0.1074 | 0.8058 $\pm$ 0.0759 | 0.7987 $\pm$ 0.0668 | 0.9022 $\pm$ 0.0426 | 0.6290 $\pm$ 0.1359 | 0.8154 $\pm$ 0.1303 |  |
| GraphSAGE                    | 2337              | 0.8174 $\pm$ 0.0636 | 0.8039 $\pm$ 0.0872 | 0.8131 $\pm$ 0.0739 | 0.8045 $\pm$ 0.0611 | 0.9034 $\pm$ 0.0407 | 0.6390 $\pm$ 0.1220 | 0.8214 $\pm$ 0.1041 |  |
| GIN                          | 3393              | 0.8240 $\pm$ 0.0708 | 0.8025 $\pm$ 0.0888 | 0.8322 $\pm$ 0.0650 | 0.8146 $\pm$ 0.0667 | 0.9008 $\pm$ 0.0418 | 0.6514 $\pm$ 0.1355 | 0.8171 $\pm$ 0.1068 |  |
| (b) Paired tests against GCN |                   |                     |                     |                     |                     |                     |                     |                     |  |
| Comparison                   | Mean $\Delta$ AUC | DeLong p            | McNemar p           | Holm p (DeLong)     | Sig. (DeLong)       | Holm p (McNemar)    | Sig. (McNemar)      |                     |  |
| GCN vs GAT                   | 0.0027            | 0.7192              | 0.05737             | 1.0                 | no                  | 0.1721              | no                  |                     |  |
| GCN vs GraphSAGE             | -0.0027           | 0.819               | 0.146               | 1.0                 | no                  | 0.292               | no                  |                     |  |
| GCN vs GIN                   | 0.001             | 0.6314              | 0.2632              | 1.0                 | no                  | 0.292               | no                  |                     |  |

TABLE S3

IDENTITY-EMBEDDING DIMENSION SWEEP ( $d \in \{0, 2, 4, 8, 16, 32\}$ ) ACROSS THREE TOPOLOGIES. MEAN  $\pm$  STD OVER 3 SEEDS  $\times$  5 FOLDS.

| id_dim | Node encoding                | Topology        | Effective rank (conv1) | ROC-AUC             | F1                  | MCC                 |
|--------|------------------------------|-----------------|------------------------|---------------------|---------------------|---------------------|
| 0      | Scalar node (paper baseline) | No graph        | 1.01                   | 0.8967 $\pm$ 0.0479 | 0.7901 $\pm$ 0.0696 | 0.6242 $\pm$ 0.1292 |
| 0      | Scalar node (paper baseline) | Corr+MST        | 1.0                    | 0.9056 $\pm$ 0.0408 | 0.7988 $\pm$ 0.0595 | 0.6320 $\pm$ 0.1218 |
| 0      | Scalar node (paper baseline) | Fully connected | 1.0                    | 0.8973 $\pm$ 0.0432 | 0.7917 $\pm$ 0.0704 | 0.6113 $\pm$ 0.1389 |
| 2      | Identity embedding (d=2)     | No graph        | 1.45                   | 0.9111 $\pm$ 0.0416 | 0.8038 $\pm$ 0.0534 | 0.6405 $\pm$ 0.1051 |
| 2      | Identity embedding (d=2)     | Corr+MST        | 1.19                   | 0.9051 $\pm$ 0.0412 | 0.8051 $\pm$ 0.0590 | 0.6429 $\pm$ 0.1170 |
| 2      | Identity embedding (d=2)     | Fully connected | 1.0                    | 0.8973 $\pm$ 0.0432 | 0.7909 $\pm$ 0.0723 | 0.6085 $\pm$ 0.1429 |
| 4      | Identity embedding (d=4)     | No graph        | 2.02                   | 0.9136 $\pm$ 0.0355 | 0.8022 $\pm$ 0.0493 | 0.6399 $\pm$ 0.1034 |
| 4      | Identity embedding (d=4)     | Corr+MST        | 1.38                   | 0.9055 $\pm$ 0.0411 | 0.8094 $\pm$ 0.0503 | 0.6488 $\pm$ 0.0966 |
| 4      | Identity embedding (d=4)     | Fully connected | 1.0                    | 0.8973 $\pm$ 0.0432 | 0.7913 $\pm$ 0.0729 | 0.6093 $\pm$ 0.1437 |
| 8      | Identity embedding (d=8)     | No graph        | 2.79                   | 0.9119 $\pm$ 0.0396 | 0.8139 $\pm$ 0.0576 | 0.6602 $\pm$ 0.1121 |
| 8      | Identity embedding (d=8)     | Corr+MST        | 1.62                   | 0.8238 $\pm$ 0.2294 | 0.7898 $\pm$ 0.0838 | 0.5670 $\pm$ 0.2503 |
| 8      | Identity embedding (d=8)     | Fully connected | 1.0                    | 0.8293 $\pm$ 0.2066 | 0.7686 $\pm$ 0.0790 | 0.5213 $\pm$ 0.2401 |
| 16     | Identity embedding (d=16)    | No graph        | 3.97                   | 0.9141 $\pm$ 0.0386 | 0.8115 $\pm$ 0.0566 | 0.6472 $\pm$ 0.1209 |
| 16     | Identity embedding (d=16)    | Corr+MST        | 2.04                   | 0.9046 $\pm$ 0.0400 | 0.7978 $\pm$ 0.0639 | 0.6256 $\pm$ 0.1357 |
| 16     | Identity embedding (d=16)    | Fully connected | 1.0                    | 0.8973 $\pm$ 0.0432 | 0.7912 $\pm$ 0.0732 | 0.6094 $\pm$ 0.1444 |
| 32     | Identity embedding (d=32)    | No graph        | 5.57                   | 0.9143 $\pm$ 0.0368 | 0.8117 $\pm$ 0.0458 | 0.6523 $\pm$ 0.0915 |
| 32     | Identity embedding (d=32)    | Corr+MST        | 2.58                   | 0.8538 $\pm$ 0.1847 | 0.7884 $\pm$ 0.0718 | 0.5902 $\pm$ 0.1960 |
| 32     | Identity embedding (d=32)    | Fully connected | 1.0                    | 0.7787 $\pm$ 0.2865 | 0.7606 $\pm$ 0.0851 | 0.5128 $\pm$ 0.2412 |

TABLE S4

INTERNAL GRAPH GAP BY IDENTITY-EMBEDDING DIMENSION. HOLM-CORRECTED ACROSS ALL 12 (DIMENSION, CONTRAST) HYPOTHESES JOINTLY.

| id_dim | Node encoding                | Contrast                   | $\Delta$ ROC-AUC | Wilcoxon p (15 paired folds) | Wins / 15 | Holm p | Significant (Holm) |
|--------|------------------------------|----------------------------|------------------|------------------------------|-----------|--------|--------------------|
| 0      | Scalar node (paper baseline) | Corr+MST – No graph        | 0.0089           | 0.0238                       | 11        | 0.2222 | no                 |
| 0      | Scalar node (paper baseline) | Corr+MST – Fully connected | 0.0083           | 0.0215                       | 11        | 0.2222 | no                 |
| 2      | Identity embedding (d=2)     | Corr+MST – No graph        | -0.006           | 0.1577                       | 4         | 0.4191 | no                 |
| 2      | Identity embedding (d=2)     | Corr+MST – Fully connected | 0.0078           | 0.0637                       | 10        | 0.3185 | no                 |
| 4      | Identity embedding (d=4)     | Corr+MST – No graph        | -0.0081          | 0.0202                       | 3         | 0.2222 | no                 |
| 4      | Identity embedding (d=4)     | Corr+MST – Fully connected | 0.0081           | 0.0408                       | 10        | 0.2471 | no                 |
| 8      | Identity embedding (d=8)     | Corr+MST – No graph        | -0.0882          | 0.0782                       | 6         | 0.3185 | no                 |
| 8      | Identity embedding (d=8)     | Corr+MST – Fully connected | -0.0056          | 0.4543                       | 10        | 0.4543 | no                 |
| 16     | Identity embedding (d=16)    | Corr+MST – No graph        | -0.0095          | 0.0256                       | 4         | 0.2222 | no                 |
| 16     | Identity embedding (d=16)    | Corr+MST – Fully connected | 0.0072           | 0.1397                       | 9         | 0.4191 | no                 |
| 32     | Identity embedding (d=32)    | Corr+MST – No graph        | -0.0605          | 0.0185                       | 2         | 0.222  | no                 |
| 32     | Identity embedding (d=32)    | Corr+MST – Fully connected | 0.0751           | 0.0353                       | 11        | 0.2471 | no                 |